ELE 537: Communication Systems

Final Exam: Solution KEY

Fall 2018

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Problem 1:

a) ⊕ $B_s \uparrow \Rightarrow \frac{S}{N} \uparrow$

b) ⊕ $K \uparrow \Rightarrow P_b = 2^{-K} \downarrow$

c) ⊕ $\frac{S}{N} \uparrow$

d) ⊕ Mod. Efficiency = 100%

e) ⊕ better performance

⊖ $B_s \uparrow \Rightarrow R \uparrow \Rightarrow$ Bandwidth ↑

⊖ overhead ↑

⊖ system complexity ↑

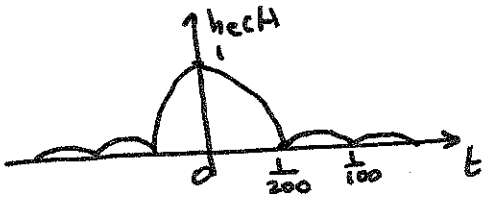
⊖ No frequency synchronisation capability
or
Envelope detectors can not be used

⊖ More complex & Local Oscillator must be synchronised.

Problem 2:

a) $B_{signal} \leq B_{channel} \Rightarrow$ No ISI
since no filtering ⇒ no pulse broadening

b)



minimum value of $T_s = \frac{1}{200}$ s

⇒ maximum value of $D = \frac{1}{T_s} = 200$ bauds.

Problem 3:

$R = 1 \times B_s = 10$ kbits/s ⇒ $B_s = 10$ kHz

$w(t) = 10 \cos(200\pi t) \Rightarrow w'(t) = -2000\pi \sin(200\pi t)$

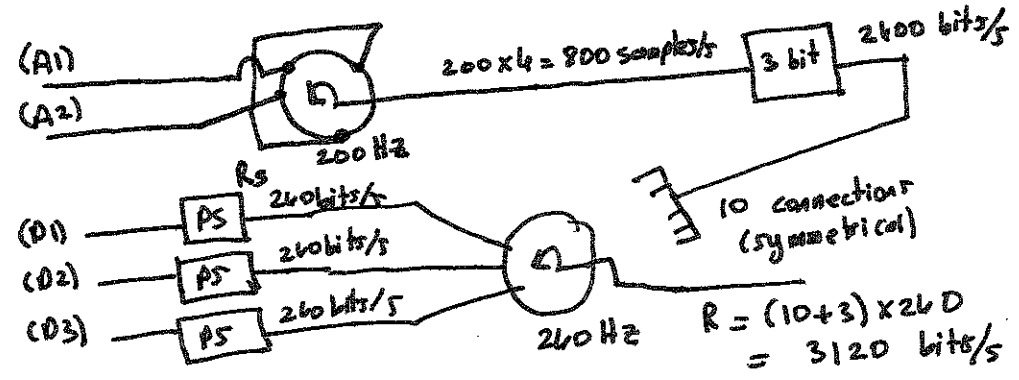
To eliminate slope-overload noise: $\delta B_s \geq \max|w'(t)|$

$\Rightarrow \delta \geq \frac{2000\pi}{B_s} = \frac{\pi}{5} = 0.6283$ V

⇒ To minimize granular noise: $\boxed{\delta = \frac{\pi}{5} = 0.6283 \text{ V}}$

Problem 4:

a)



$2400 = k R_s$

$R_s \geq 235$

$\Rightarrow k \leq \frac{2400}{235} = 10.2$

$\Rightarrow k = 10$

$\Rightarrow R_s = 240$ bits/s

$R = (10+3) \times 240 = 3120$ bits/s

b) $D = R = 3120$ bauds (binary)

$$D = \frac{2B}{1+r} \Rightarrow B = \frac{1.5 \times 3120}{2} = 2340 \text{ Hz}$$

Problem 5:

a) $g(t) = A_c\, m(t) = 10 \cos(200\pi t)$

$$\langle s^2(t) \rangle = \frac{1}{2} \langle |g(t)|^2 \rangle = \frac{100}{2 \times 2} = \frac{50}{2} W = 25 W$$

b) $g(t) = A_c\, e^{j\theta(t)}$

$$\Rightarrow \langle s^2(t) \rangle = \frac{1}{2} \langle |g(t)|^2 \rangle = \frac{A_c^2}{2} = \frac{100}{2} = 50 W$$

c) $g(t) = A_c [1 + m(t)]$

$$\Rightarrow \langle s^2(t) \rangle = \frac{1}{2} \langle A_c^2 + 2A_c^2 m(t) + A_c^2 m^2(t) \rangle$$

$$= \frac{A_c^2}{2} + A_c^2 \underbrace{\langle m(t) \rangle}_{=0} + \frac{A_c^2}{2} \underbrace{\langle m^2(t) \rangle}_{=1/2} = A_c^2 \times \frac{3}{4} \quad \frac{100}{4} W = 75 W$$

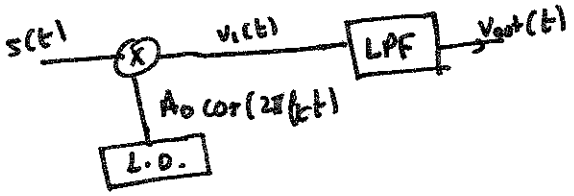
d) $g(t) = A_c [m(t) + j \hat{m}(t)]$ where $\hat{m}(t) = \cos(200\pi t - \pi/2) = \sin(200\pi t)$

$$\rightarrow g(t) = 10 [\cos(200\pi t) + j \sin(200\pi t)] = 10 e^{j 200\pi t} \Rightarrow |g(t)| = 10$$

$$\rightarrow \langle s^2(t) \rangle = \frac{1}{2} \langle |g(t)|^2 \rangle = \frac{100}{2} = 50 W$$

Problem 6:

a) $s(t) = A_c [1 + m(t)] \cos(\omega_c t)$



$$v_1(t) = s(t) \cos(\omega_c t)$$

$$= A_c [1 + m(t)] \cos^2(\omega_c t)$$

$$= \frac{A_c}{2} [1 + m(t)] + \frac{A_c}{2} [1 + m(t)] \cos(2\omega_c t)$$

(LPF)

$\rightarrow V_{out}(t) = \frac{A_c}{2} + \frac{A_c}{2} m(t)$ Linear function of $m(t)$

$\Rightarrow$ detection successful.

b) $V_{out}(t) = K R(t)$

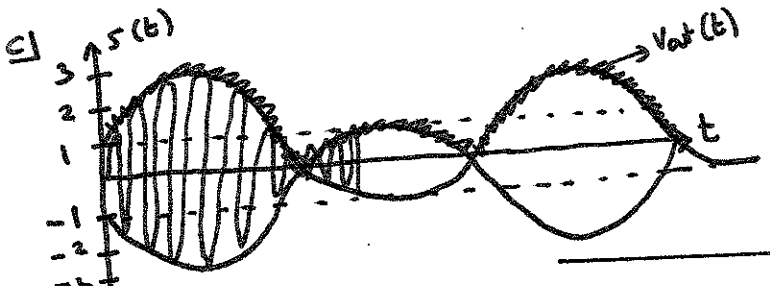
$$= K |g(t)| = K A_c |1 + m(t)|$$

since $1 + m(t)$ can be +ve or -ve

$\Rightarrow V_{out}(t) = K A_c |1 + m(t)|$ non linear function of $m(t)$

$\Rightarrow$ detection not successful

c)

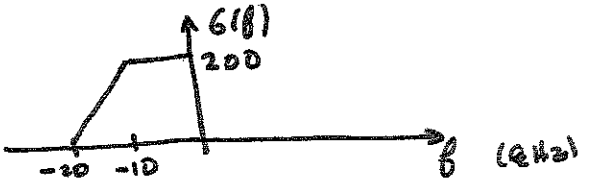


$A_{min} = A_c(1-2) = -A_c = -1$

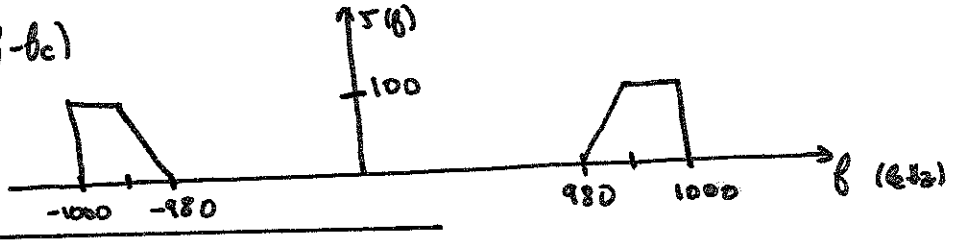
$A_{max} = A_c(1+2) = 3A_c = 3$

Problem 7:

$$G(f) = \begin{cases} 2A_c M(f) & f<0 \\ 0 & f>0 \end{cases}$$



$$S(f) = \frac{1}{2} G(f-f_c) + \frac{1}{2} G^*(-f-f_c)$$



Problem 8:

a) $y(t) = \cos^3(\omega_c t) = \cos^2(\omega_c t) \cdot \cos(\omega_c t)$

$$= \left[\frac{1}{2} + \frac{1}{2}\cos(2\omega_c t)\right]\cos(\omega_c t)$$

$$= \frac{1}{2}\cos(\omega_c t) + \frac{1}{2}\cos(2\omega_c t)\cos(\omega_c t)$$

$$= \frac{1}{2}\cos(\omega_c t) + \frac{1}{4}\cos(\omega_c t) + \frac{1}{4}\cos(3\omega_c t)$$

$$= \underbrace{\frac{3}{4}}_{a_1}\cos(\omega_c t) + \underbrace{\frac{1}{4}}_{a_3}\cos(3\omega_c t)$$

$$THD = \sqrt{\frac{\sum_{n=2}^{\infty} a_n^2}{a_1^2}} \times 100\% = \frac{a_3}{a_1} \times 100\% = 33.33\%$$

Problem 9:

$$s(t) = \mathrm{Re}\left\{ \underbrace{\left[5e^{-j\omega_a t} + 5e^{j\omega_a t}\right]}_{g(t)} e^{j\omega_c t} \right\}$$

$$\Rightarrow g(t) = 10\cos(\omega_a t)$$

$$PEP = \frac{1}{2}\max|g(t)|^2 = \frac{100}{2} = 50\,W$$

Problem 10:

a) $f_{LO} = f_c + f_{IF} = 100 + 10 = 110$ MHz

b) $f_{image} = f_c + 2f_{IF} = 120$ MHz

c) IF BPF selective in the range $\left[f_{IF} - \frac{B}{2} \quad f_{IF} + \frac{B}{2}\right] = [9.9 \quad 10.1]$ MHz

Problem 11:

a) $P_{av} = \frac{\langle s^2(t)\rangle}{50} = \frac{(100)^2}{50} = 100\,W$

b) PM: $\theta(t) = \frac{\Delta\rho}{100} m(t) = 4\cos(300\pi t) \Rightarrow m(t) = 0.04\cos(300\pi t)$

c) FM: $\theta(t) = D_f \int_{-\infty}^{t} m(\tau)d\tau$

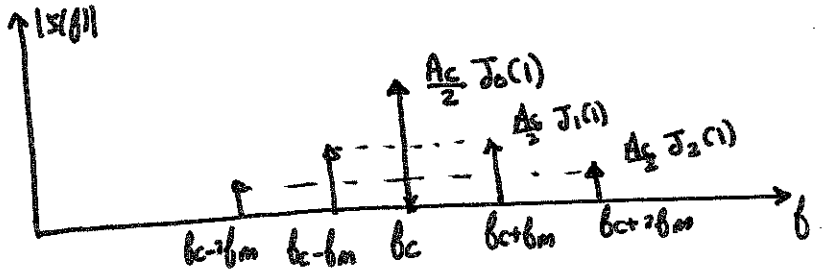
$\Rightarrow \frac{d\theta(t)}{dt} = D_f\, m(t) \Rightarrow m(t) = \frac{1}{D_f}\frac{d\theta(t)}{dt} = \frac{1}{10^4}\left(-1200\pi \sin(300\pi t)\right)$

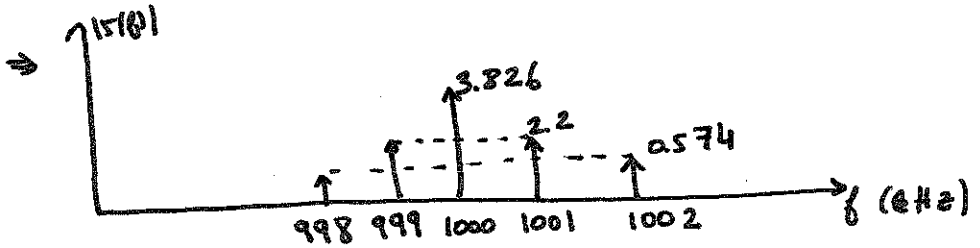
$\Rightarrow m(t) = -0.377 \sin(300\pi t)$

Problem 12:

$$\begin{cases} g(t) = 10\, e^{j \sin(2000\pi t)} \\ f_c = 1\ MHz = 1000\ kHz \end{cases} = A_c\, e^{j\beta \sin(\omega_m t)} \Rightarrow \begin{cases} A_c = 10 \\ \beta = 1 \\ f_m = 1\ kHz \end{cases}$$

$B = 2(\beta+1) f_m = 4\ kHz$





Problem 13:

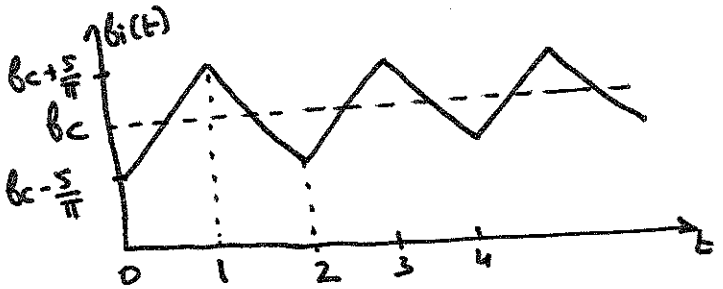
$$B < \frac{1}{2\pi RC} < f_c \Rightarrow \frac{1}{2\pi f_c} < RC < \frac{1}{2\pi B} \quad \text{where } f_c = 1\ MHz \text{ \& } B = 5\ kHz$$

$$\Rightarrow 1.59 \times 10^{-7} < RC < 3.18 \times 10^{-5}$$

Problem 14:

a) $f_i(t) = \frac{1}{2\pi}\frac{d}{dt}\left[2\pi f_c t + 5\int_{-\infty}^{t} m(\tau)d\tau\right]$

$= f_c + \frac{5}{2\pi} m(t)$



b) From the figure: $\Delta F = \frac{5}{\pi} = 1.5915\ Hz$

or $\Delta F = \frac{D_f V_p}{2\pi} = \frac{5 \times 2}{2\pi} = \frac{5}{\pi} = 1.5915\ Hz$

c) Figure (c): Instaneous frequency increases from $t = 0s$ to $t = 1s$

Maximum value at $t = 1, 3, 5$

Minimum value at $t = 0, 2, 4$